

The empirical recurrence for OEIS A282836: recovering a conjecture that is not written down

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Abstract

OEIS A282836 records “Empirical recurrence of order 90”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 231$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 90, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 90 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A282836 is “Number of nX6 0..1 arrays with no 1 equal to more than one of its king-move neighbors, with the exception of exactly two elements.”. It has offset 1 and begins

5, 144, 6472, 151084, 2839946, 54433534, 957808843, 16264638662, ...

The entry states:

Empirical recurrence of order 90 (see link above)

As of the “Last modified” line on the live entry (Feb 22 2017 11:21:29, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2296$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 231$ of the 2296, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 231$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 462$ exact terms of the model returns order 90 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{90} c_i a(n-i),$$

c_1	12	c_{24}	−104143731573027820	c_{47}	11615603648778716906352	c_{70}	95221913773
c_2	189	c_{25}	248165943445038315	c_{48}	9868891941993344737973	c_{71}	38719140263
c_3	−11	c_{26}	−797144950249934667	c_{49}	11483512706861860079661	c_{72}	−3231581987
c_4	−28701	c_{27}	2002868339678572410	c_{50}	−16673036780969155377495	c_{73}	−3228103527
c_5	−197718	c_{28}	−4116306217405131990	c_{51}	−23208022769976786326574	c_{74}	1618972749
c_6	406720	c_{29}	10356721317105303015	c_{52}	−18621289726925670438420	c_{75}	14169277463
c_7	17146848	c_{30}	−20082066337034433889	c_{53}	21510738463619605164444	c_{76}	4529916003
c_8	110389992	c_{31}	34959978514371387345	c_{54}	37422322232457408757474	c_{77}	−349615356
c_9	258454441	c_{32}	−75730173243361645686	c_{55}	22619414313847100534439	c_{78}	−265802473
c_{10}	−1304031339	c_{33}	116390082313025205676	c_{56}	−27451739190868049433285	c_{79}	1944140172
c_{11}	−10050432591	c_{34}	−177884066730970789074	c_{57}	−46029238472482296735166	c_{80}	777788092
c_{12}	−30151107025	c_{35}	359860625891283407298	c_{58}	−18076797416493560629665	c_{81}	1828083163
c_{13}	40453631823	c_{36}	−422900833463368098482	c_{59}	34338430283090236582008	c_{82}	−112738923
c_{14}	350090641326	c_{37}	606111184872355670955	c_{60}	44088641737938209720557	c_{83}	−63659132
c_{15}	1681508624009	c_{38}	−1227325815451153402509	c_{61}	5023821402397012248936	c_{84}	11866572
c_{16}	−646755785400	c_{39}	946690278365050242132	c_{62}	−37203383751992766903591	c_{85}	83460041
c_{17}	−1283652984204	c_{40}	−1510148480361820259322	c_{63}	−30305993987437236261716	c_{86}	17841659
c_{18}	−75789595856422	c_{41}	3216769587663883237419	c_{64}	9117590310808246697868	c_{87}	−2534197
c_{19}	75670590709656	c_{42}	−927080286924065296125	c_{65}	29959985693139881237694	c_{88}	−1546574
c_{20}	−465962071522548	c_{43}	3108391334626172321706	c_{66}	11298523170004080189678	c_{89}	−2277022
c_{21}	2998724687083589	c_{44}	−6757273053358648755888	c_{67}	−14286583507928501904855	c_{90}	−115558
c_{22}	−6438262487435247	c_{45}	−1812643531897291078442	c_{68}	−15382153009639544204622		
c_{23}	29447315963476605	c_{46}	−6069235828481414103945	c_{69}	1043436502162504507007		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 90, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $90 + 4$ terms removed returns order 90 as well, so $\deg q_{\min} = 90 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $90 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 90 that this sequence satisfies — and that family turns out to have exactly one member.

References

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